Exercise Report

Goal

I created a systemd-based solution checks for all `pdf` files inside `~/Downloads` directory and moves them to `$HOME/archive` every 2 minutes.

Step 1: Create the script

First, I created a bash script that checks for PDF files inside `~/Downloads` and moves them to `$HOME/archive`.

File:

```
nano transfer_pdfs.sh
```

Script content:

```
#!/bin/bash
mkdir -p "$HOME/archive"
find "$HOME/Downloads" -maxdepth 1 -type f -name "*.pdf" -exec mv {}
"$HOME/archive/" \;
```

After that, I made it executable, then check it's permissions:

```
chmod +x transfer_pdfs.sh
ls -all transfer_pdfs.sh
```

```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ nano transfer_pdfs.sh

┌──(crypto-env)─(noor㉿elif)-[~]
└─$ chmod +x transfer_pdfs.sh

┌──(crypto-env)─(noor㉿elif)-[~]
└─$ ls -all transfer_pdfs.sh
-rwxrwxr-x 1 noor noor 126 Mar 11 18:37 transfer_pdfs.sh
```

Step 2: Create the systemd service file

Then I created a service file:

```
sudo nano /etc/systemd/system/transfer_pdfs.service
```

Service content:

```
[Unit]
Description=Move PDF files from Downloads to archive

[Service]
Type=oneshot
ExecStart=$HOME/transfer_pdfs.sh
```

```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ sudo nano /etc/systemd/system/transfer_pdfs.service
[sudo] password for noor:
```

Step 3: Create the systemd timer file

Since I needed the task to run every 2 minutes, I used a **systemd timer**.

```
sudo nano /etc/systemd/system/transfer_pdfs.timer
```

Timer content:

```
[Unit]
Description=Run transfer_pdfs service every 2 minutes

[Timer]
OnBootSec=2min
OnUnitActiveSec=2min
Unit=transfer_pdfs.service

[Install]
WantedBy=timers.target
```

```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ sudo nano /etc/systemd/system/transfer_pdfs.timer
```

Step 4: Reload systemd and enable the timer

After creating both files, I reloaded systemd and enabled the timer:

```
sudo systemctl daemon-reload
sudo systemctl enable --now transfer_pdfs.timer
```

```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ sudo systemctl daemon-reload

┌──(crypto-env)─(noor㉿elif)-[~]
└─$ sudo systemctl enable --now transfer_pdfs.timer
Created symlink '/etc/systemd/system/timers.target.wants/transfer_pdfs.timer' → '/etc/systemd/sy
stem/transfer_pdfs.timer'.
```

Step 5: Check that it is working

To verify everything, I checked the timer status:

```
sudo systemctl status transfer_pdfs.timer
```

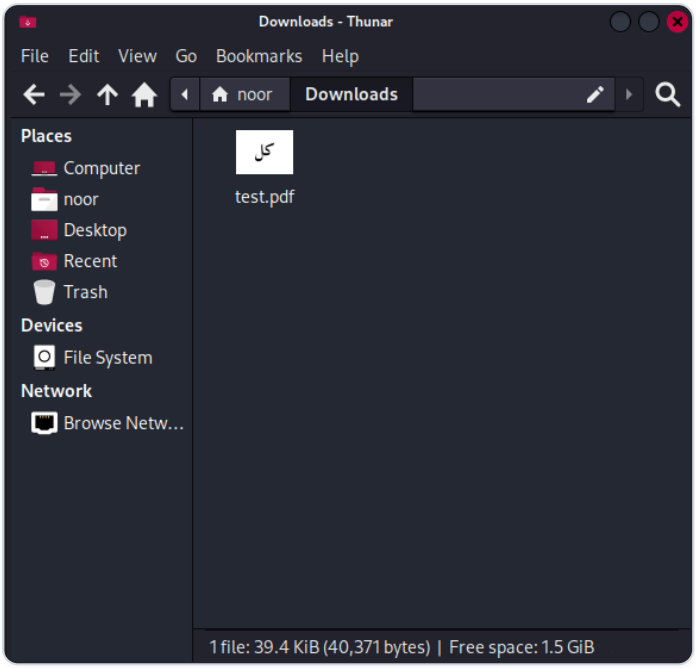
```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ sudo systemctl status transfer_pdfs.timer
● transfer_pdfs.timer - Run transfer_pdfs service every 2 minutes
     Loaded: loaded (/etc/systemd/system/transfer_pdfs.timer; enabled; preset: enabled)
     Active: active (waiting) since Wed 2026-03-11 19:25:56 UTC; 19min ago
 Invocation: 4bb0aec703604c78904ffb34b66f8f9e
    Trigger: Wed 2026-03-11 19:47:24 UTC; 1min 40s left
   Triggers: ● transfer_pdfs.service

Mar 11 19:25:56 elif systemd[1]: Started transfer_pdfs.timer - Run move-pdfs service every 2 minutes.
```

I also checked the service logs if needed:

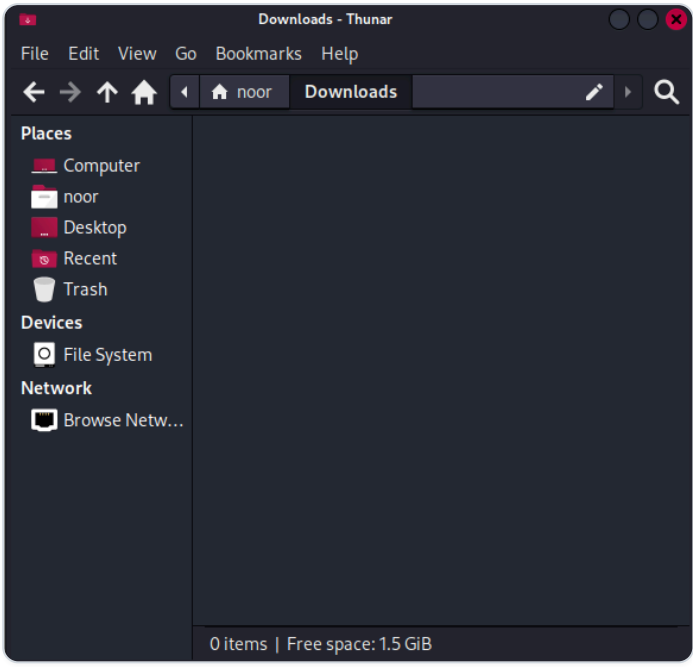
```
journalctl -u transfer_pdfs.service
```

```
┌──(crypto-env)─(noor㉿elif)-[~]
└─$ journalctl -u transfer_pdfs.service
Mar 11 19:45:24 elif systemd[1]: Starting transfer_pdfs.service - Move PDF files from Downloads to archive...
Mar 11 19:45:24 elif systemd[1]: transfer_pdfs.service: Deactivated successfully.
Mar 11 19:45:24 elif systemd[1]: Finished transfer_pdfs.service - Move PDF files from Downloads to archive.
Mar 11 19:47:35 elif systemd[1]: Starting transfer_pdfs.service - Move PDF files from Downloads to archive...
Mar 11 19:47:35 elif systemd[1]: transfer_pdfs.service: Deactivated successfully.
Mar 11 19:47:35 elif systemd[1]: Finished transfer_pdfs.service - Move PDF files from Downloads to archive.
```

Downloads directory before and after

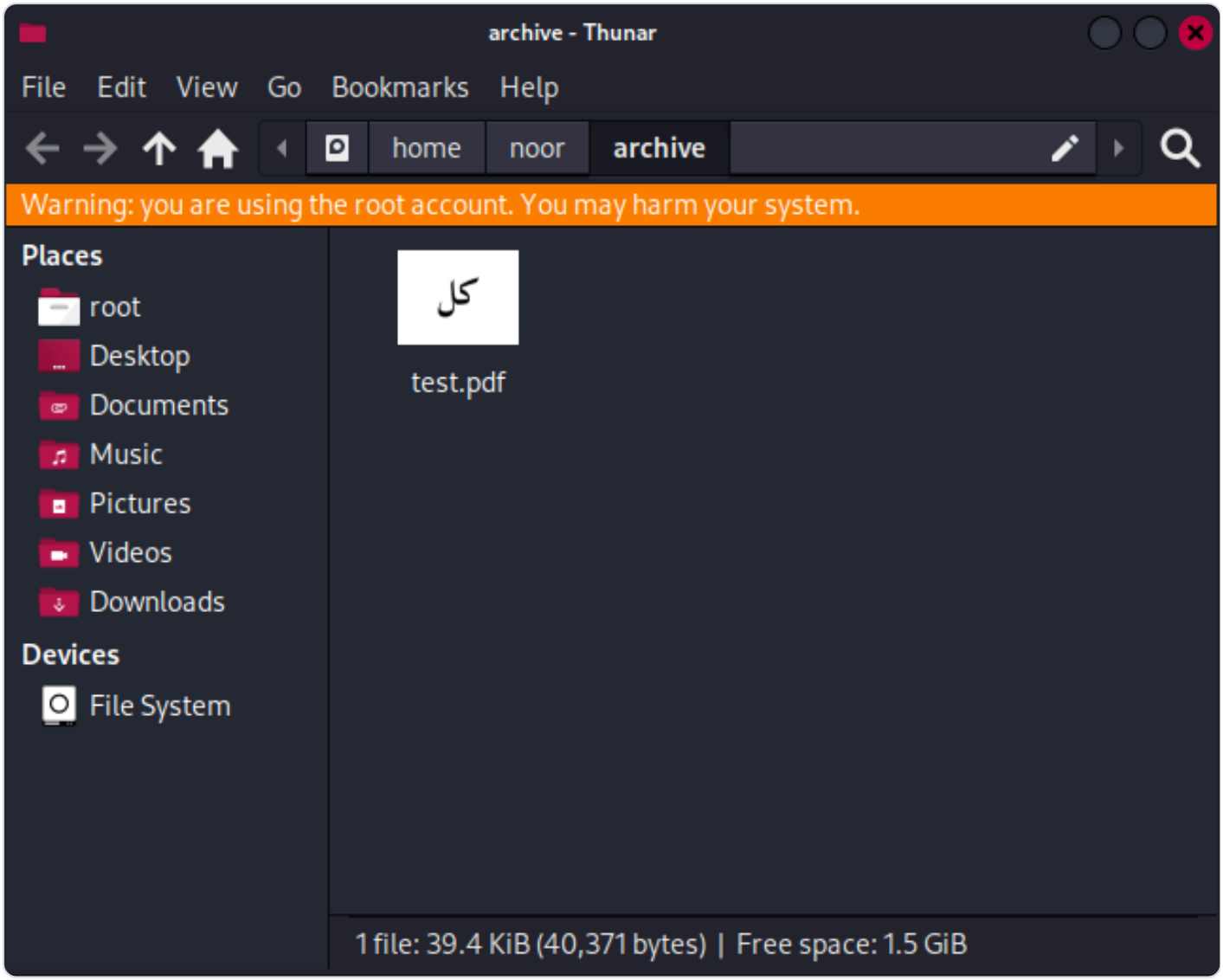


Downloads directory before



Downloads directory after

Arcive directory after transferring



How the solution works

- The script checks `~/Downloads` for any file ending with `.pdf`.
- If it finds any PDF files, it moves them to `$HOME/archive`.
- The timer runs the service every 2 minutes.
- This avoids using cron jobs and uses systemd as the alternative scheduler.

Files I created

1) Script file

`transfer_pdfs.hs`

2) Service file

`/etc/systemd/system/transfer_pdfs.service`

3) Timer file

`/etc/systemd/system/transfer_pdfs.timer`

Final commands summary

```
nano transfer_pdfs.sh
chmod +x transfer_pdfs.sh
ls -all transfer_pdfs.sh
sudo nano /etc/systemd/system/transfer_pdfs.service
sudo nano /etc/systemd/system/transfer_pdfs.timer
sudo systemctl daemon-reload
sudo systemctl enable --now transfer_pdfs.timer
sudo systemctl status transfer_pdfs.timer
journalctl -u transfer_pdfs.service
```

Conclusion

I solved the task by using a **bash script + systemd service + systemd timer**. The timer runs every 2 minutes, checks for PDF files in `~/Downloads`, and moves them to `$HOME/archive` without using cron jobs.